Power

Power is the rate at which work is done. It is the work/time ratio.

$$\text{Power} = \frac{\text{Work}}{\text{time}}$$

The standard metric unit of power is the **Watt**. Sometimes *horsepower* is used to describe the power delivered by a machine.

The more work done in a shorter period of the time the more power.

All machines have a power rating. The power rating indicates the rate at which that machine can do work

A person is also a machine that has a *power rating*. Some people are more power-full than others.

1. Two students, Will and Ben, are weightlifting. Will lifts the 100kg barbell over his head 10 times in 60 seconds; Ben lifts the 100kg barbell over his head 10 times in 10 seconds. Who delivers the most power? _____ Explain your answers.

2. John pushes a shopping trolley doing 500J work in 10 seconds. What is the power?

3. Jan pushes her broken down car for 5 seconds and does 300J work. What was the power involved?

PUTTING TWO IDEAS TOGETHER

If we know the equation for finding work and the equation for finding power we can combine the two ideas.

1. Tom pushes a cart for 10 seconds with a force of 120N over a distance of 10M.

a. How much work has he done?

b. What was the power involved?

2. Jim pushes a rock for 20 seconds with a force of 100N over a distance of 20M.

a. How much work has he done?

b. What was the power involved?

3. Tim drags a bag of potatoes for 5 seconds with a force of 150N over a distance of 5M.

a How much work has he done?

b What was the power involved?

4. Rod pushes a wheel barrow for 10 seconds with a force of 50N over a distance of 100M.

a How much work has he done?

b What was the power involved?

5. Tim pushes a box for 5 seconds with a force of 10N over a distance of 10M.

a How much work has he done?

b What was the power involved?

KW/h Passenger/speed= Energy-per-km	**Boeing 747** jumbo jet	**Queen Marry** or large ocean liner	**SUV** or large car	**Bicycle**	**Person** on foot
Weight	369 tons (fully loaded)	81,000 tons	2.5 tons	100 kg with person	80kg (176 lb)
Cruising speed	900km/h (560 mph)	52km/h (32mph)	100km/h (62mph)	20km/h (12.5mph)	5km/h (3.1mph)
Maximum power	77,000kW (100,000hp)	120,000kW (160,000hp)	200kW (275hp)	2000 W (professional)	2000 W
Energy at cruising	65,000kW (87,000hp)	90,000 kW (120,000hp)	130 kW (174hp)	80 W (0.1hp)	280 W (0.38 hp)
Passengers	450	3000	4	1	1
Power/passenger	140 kW	40kW	50kW	80 W	280 W
Energy/passenger/km	580 kilo joule*	2800 kilo joule*	1800 kilo joule*	14.4 kilo joule*	200 kilo joule*